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PROJECT REPORT ON

"SOUL GAMBLE: Data-Structure Driven Psychological Poker & Stand Showdown Simulator"

Subject: Data Structures - Lab

Submitted By:

Name	Branch	PRN
Divyansh Motwani	CSE	25070721015

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1. ACKNOWLEDGEMENT

We take this opportunity to express our heartfelt gratitude to everyone who has guided and supported us throughout the completion of our project titled "SOUL GAMBLE: Data-Structure Driven Psychological Poker & Stand Showdown Simulator."

First and foremost, we extend our deepest thanks to Dr. Rakesh Salakapuri, our professor and project guide, for his constant encouragement, valuable guidance, and continuous support. His insightful feedback and in-depth knowledge helped us understand the technical intricacies of linear and non-linear data structures—including Double-Ended Queues (Deques), LIFO Stacks, Hash Maps, and Finite State Machines—and bring our concept to life as a fully functional interactive prototype. His mentorship has been a constant source of inspiration throughout this journey.

We would like to express our sincere gratitude to Dr. Rajinikanth Aluvalu, Director of Symbiosis Institute of Technology, Hyderabad, for providing us with the necessary infrastructure, resources, and an environment that encourages innovation, experimentation, and research. His vision for academic excellence has motivated us to pursue this project with dedication and creativity.

Finally, we would like to express our gratitude to our peers and classmates for their helpful suggestions, cooperation, and moral support throughout the project. We are also thankful to Symbiosis Institute of Technology, Hyderabad, for providing us with the platform and resources to transform our ideas into reality.

2. ABSTRACT AND INTRODUCTION

Soul Gamble is an experimental, interactive game system designed to simulate high-stakes 5-Card Draw Poker and psychological warfare using core computer science data structures and behavioral state modeling. Inspired by the iconic showdown between Jotaro Kujo and Daniel J. D'Arby, the card deck is modeled as a Double-Ended Queue (Deque) to support both fair top-dealing and sleight-of-hand bottom or second-dealing, while the wagered soul chips are managed via a Last-In, First-Out (LIFO) Stack. Furthermore, the opponent AI is governed by a Finite State Machine (FSM) driven by dynamic Composure and Paranoia metrics.

This project bridges classical data structure operations and interactive game AI, providing a novel approach to visualizing how real-time stack, deque, and state-transition manipulations govern complex turn-based decision systems.

Traditional digital card games rely on simple randomized arrays, which fail to capture the physical mechanics of card manipulation, multi-tier collateral wagering, and psychological intimidation. Computational models built on specialized data structures now allow for transparent, inspectable simulations of both physical sleight-of-hand and cognitive pressure.

The Soul Gamble framework treats every playing card, soul chip, and psychological tell as a discrete data node. By leveraging a Deque for deck manipulation, a LIFO Stack for bidirectional soul chip wagers, Hash Maps for linear-time frequency-based hand evaluation, and a Finite State Machine (FSM) for opponent cognition, the system mimics the strategic and psychological tension of a real-time showdown. This approach reinforces core Data Structures concepts through live visual telemetry ("DS Inspector") and offers practical insights into adversarial game AI design and interactive probability visualization.

3. METHODOLOGY & BIDIRECTIONAL CHIP ARCHITECTURE

1. Deck Initialization & Deque Construction: Create a standard 52-card dataset with rank (2–14) and suit attributes. Store the shuffled deck inside a Double-Ended Queue (Deque) supporting constant-time extraction from the front (popFront for normal top deals), second index (extractSecond for second deals), and rear (popBack for rigged bottom deals).
2. LIFO Soul Pot Stack & Bidirectional Chip Transfers: Represent every wagered soul chip (Joseph, Polnareff, Jotaro, Avdol, Kakyoin, and Holly) as an owner-tagged object node pushed onto the central LIFO Soul Pot Stack.

When Jotaro Wins a Pot: All nodes are popped from the LIFO Pot Stack. Original D'Arby soul chips (owner: JOSEPH or POLNAREFF) are routed into Jotaro's locked, non-wagerable Liberated Souls Vault (0/12) so recovered friends can never be risked again. Any recovered JOTARO, AVDOL, or KAKYOIN chips return directly to Jotaro's Active Wagerable Bank.

When D'Arby Wins a Pot (or Jotaro Folds): All nodes in the LIFO Pot Stack—including Jotaro's lost plain chips and pledged Crusader chips—transfer directly into D'Arby's active bank without any artificial 12-chip cap. D'Arby displays captured Jotaro chips in his bank HUD and actively wagers those captured chips first (in LIFO order) in subsequent hands, allowing Jotaro to win them back.

3. Kinetic Vision QTE & Deck Tampering: Provide a 5.0-second real-time dealing window where the player can trigger Star Platinum via the Spacebar to inspect the Deque extraction pointer. If a cheat is caught, the player can either publicly expose D'Arby (stationing the persistent Neutral Dealer Boy beside the table for all remaining hands) or pocket the intel to perform a covert Deck Tamper swap during the Draw phase.
4. Finite State Machine (FSM) Behavioral Modeling: Track D'Arby's cognitive state across five FSM nodes: SMUG, CALCULATING, GREEDY, PARANOID, and CATATONIC. State transitions are driven by Composure (0–100%) and Paranoia (clamped at a 10% minimum floor up to 100%), scaled by modifiers such as Blind Hand locking (1.5× Paranoia multiplier).
5. Hash-Map Hand Evaluation & Match Termination: Evaluate 5-card poker hands using a Rank and Suit Frequency Hash Map to classify hands from High Card up to Royal Flush. The match concludes immediately when Jotaro recovers all 12 original D'Arby soul chips (or triggers a 100% Paranoia collapse via DIO's Stand Secret), or ends in Defeat if Jotaro exhausts all active and vault chips (or loses a pledged Holly Kujo chip).

4. PSYCHOLOGICAL WARFARE & GAME THEORY IN IMPERFECT-INFORMATION POKER

Five-Card Draw Poker is a foundational imperfect-information game in game theory, where players make optimal decisions under uncertainty by combining mathematical probability with psychological deception. In Soul Gamble, the game extends beyond card odds into cognitive warfare, where breaking the opponent's mental composure is as viable a victory condition as holding the strongest five-card hand.

Mirroring hierarchical cognitive models, the simulator divides table state into two analytical layers:

Observable Table Layer (Manifest State): The publicly visible actions on the casino felt—such as Street Bets (1 or 2 Chips), checking, raising, and the number of cards discarded during the Draw Phase (0 to 3 cards). In our model, this corresponds to the public UI state, the LIFO Soul Pot size, and D'Arby's visible Discard Tells (discarding 0 cards signals a pat Straight/Flush or snow bluff; discarding 1 card signals Two Pair or a four-card draw; discarding 2 cards signals Three of a Kind; discarding 3 cards signals One Pair).

Hidden Cognitive & Rigged Layer (Latent State): The unobservable mechanics operating beneath the surface—such as D'Arby's bottom-dealt cards at the tail of the Deque, Jotaro's unlooked Blind Hand, and D'Arby's internal fear of Star Platinum's speed. Computationally, this corresponds to the hidden Deque pointers, the Bluff Probability calculator in Muhammad Avdol's Adviser modal, and the internal Composure and Paranoia meters.

The Role of Deception, Bluffing, and Cognitive Collapse

In standard poker, a player holding a weak hand folds to heavy aggression. However, when psychological pressure is applied—such as locking a hand Blind (1.5× Paranoia multiplier), pledging allied souls from the Soul Vault, or executing an intimidating Bluff Raise—the opponent's rational evaluation fractures. As D'Arby's internal Paranoia (P) rises, each percentage point directly increases the probability (P%) that demanding DIO's Stand Secret will trigger a total cognitive overload (CATATONIC state), forcing D'Arby to forfeit even when holding a rigged hand.

5. CORE DATA STRUCTURES & FINITE STATE MACHINES (FSM)

A Double-Ended Queue (Deque), a LIFO Stack, and a Finite State Machine (FSM) form the mathematical backbone of Soul Gamble. In a Deque, card nodes are inserted or removed from both the front (top of the deck) and the rear (bottom of the deck) in constant O(1) time complexity:

Fair Deal → Deque.popFront() | Second Deal Cheat → Deque.extractSecond() | Bottom Deal Cheat → Deque.popBack()

In the Finite State Machine (FSM), the AI opponent exists in exactly one behavioral state at any given moment, transitioning from state Si to state Sj based on current psychological telemetry (Composure C and Paranoia P, where P ≥ 10%).

State Transition & Action Probability Matrix

During any betting street, D'Arby's decision weights for executing an Aggressive Raise/Bluff, a Cautious Call/Check, or a Fold/Mental Collapse depend directly on his active FSM node:

Current AI State \ Action	Aggressive Raise / Bluff	Cautious Call / Check	Fold / Mental Collapse
SMUG (C ≥ 70%, P < 30%)	0.65 (65%)	0.30 (30%)	0.05 (5%)
CALCULATING (45% ≤ C < 70%, 30% ≤ P < 60%)	0.35 (35%)	0.50 (50%)	0.15 (15%)
PARANOID (C < 45%, 60% ≤ P < 100%)	0.15 (15%)	0.45 (45%)	0.40 (40%)
CATATONIC (P = 100% — Breakdown)	0.00 (0%)	0.00 (0%)	1.00 (100% Instant Forfeit)

Additionally, the Demand DIO's Stand Secret confrontation mechanic operates on a direct linear probability rule where every 1% of Paranoia yields a 1% chance of triggering an immediate psychological collapse:

P(Instant Psychological Collapse) = Paranoia (%) ÷ 100

6. APPLICATIONS AND FUTURE SCOPE

Applications

1. Interactive Data Structure Visualization: Providing an engaging, gamified sandbox (DS Inspector) to observe real-time Deque, LIFO Stack, Hash Map, and FSM state operations.
2. Adversarial Game AI & Behavioral Modeling: Demonstrating how emotional telemetry (Composure and Paranoia) dynamically alters AI decision weights in imperfect-information games.
3. Probability & Game Theory Education: Teaching conditional probability, pot odds, discard tells, and risk-reward collateral management through Muhammad Avdol's strategic adviser module.
4. Algorithmic Sleight-of-Hand Simulation: Modeling non-standard queue pointer manipulations (bottom dealing, second dealing, and in-place node swapping via Star Platinum's Deck Tamper).
5. Bidirectional Token Accounting: Tracking multi-owner chip inventories across active wagerable banks, opponent captured stacks, and locked non-wagerable vaults.

Future Scope

- Multiplayer Networked Showdowns: Extending the state engine via WebSockets so two human players can engage in real-time mind-games, secret deck-tampering QTEs, and bluffing duels.
- Monte Carlo Tree Search (MCTS) AI Upgrade: Combining the psychological Finite State Machine with a Monte Carlo simulation engine for deeper multi-street poker lookahead.
- Expanded Stand Battles & Custom Data Structures: Adding new opponents (such as Telence T. D'Arby) with unique data-structure-based mechanics (Binary Search Trees and Priority Queues).

7. SYSTEM PROTOTYPE & LIVE TELEMETRY SHOWCASE

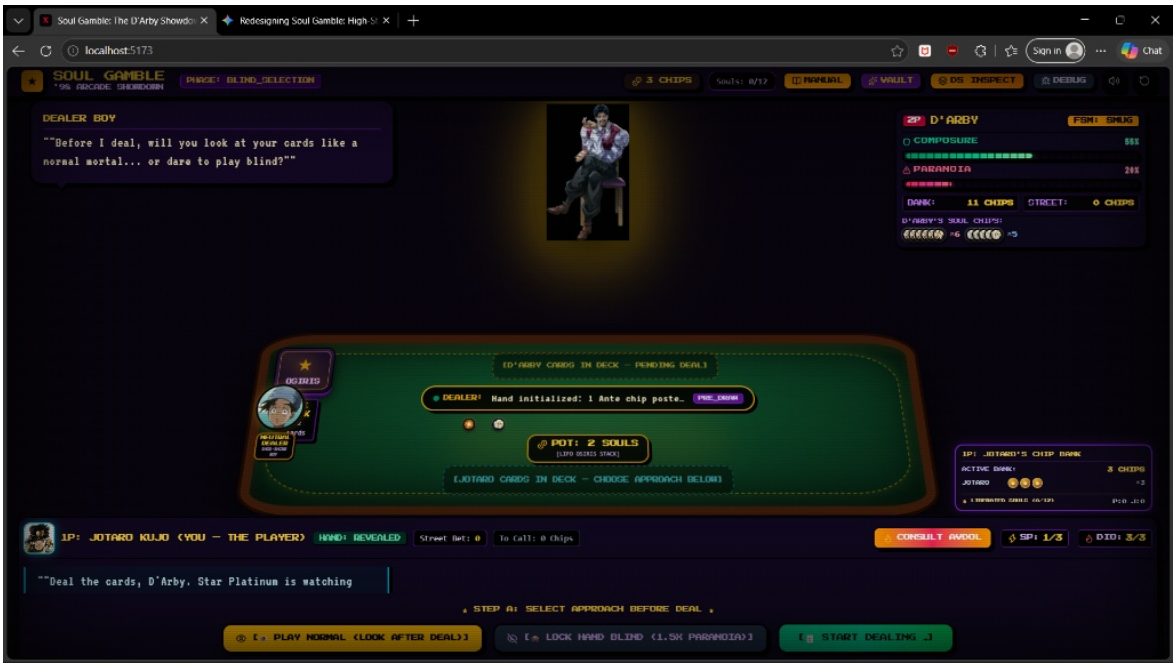


Figure 1: Main 90s Arcade FPV Poker Table prototype showing D'Arby's Composure/Paranoia gauges, circular Joseph and Polnareff soul chips, the 3D Osiris Deck Deque with the Neutral Dealer Boy on the left, the central LIFO Soul Pot Stack, and Jotaro's Chip Bank on the right.

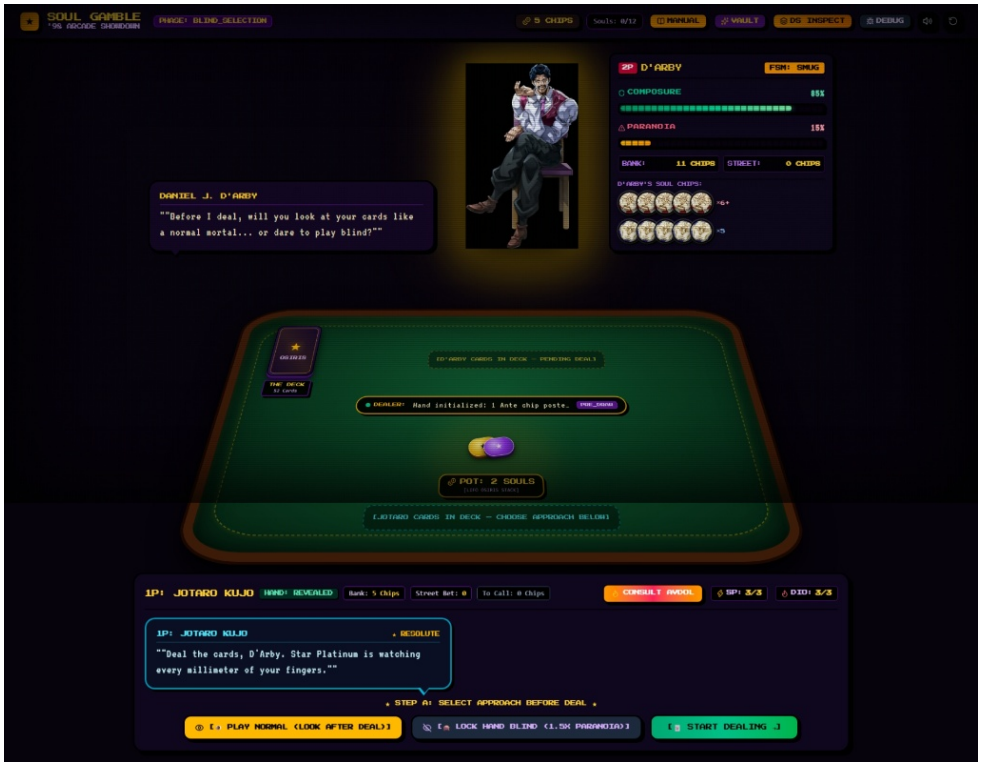


Figure 2: Pre-Deal Setup phase where the player selects between Normal Play or Locking the Hand Blind (1.5× Paranoia Multiplier) before initiating the 5.0-second dealing animation.

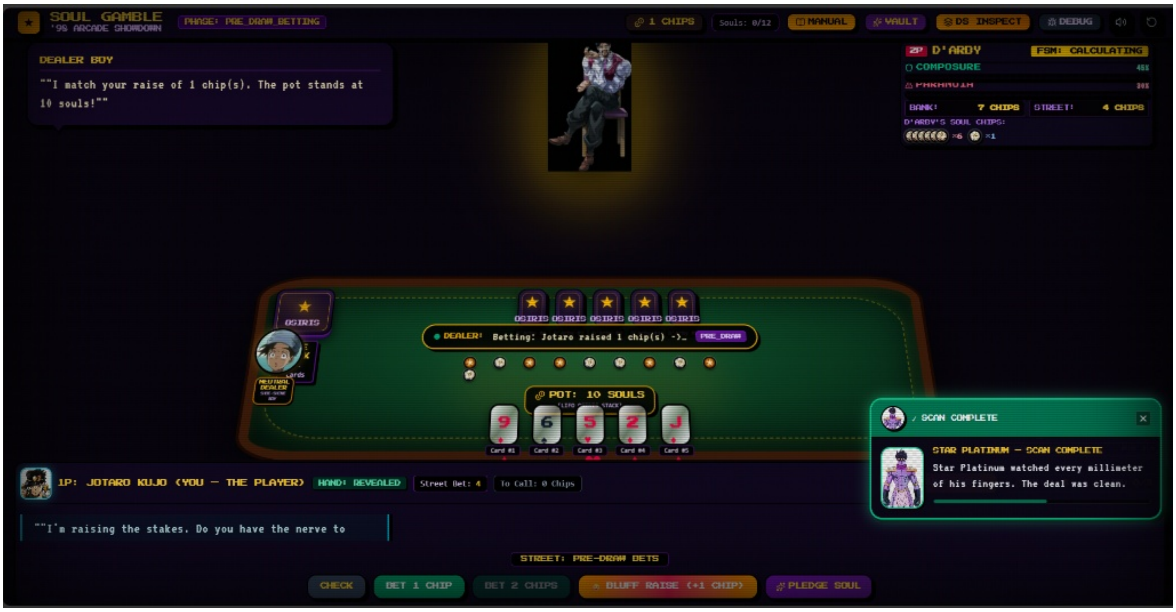


Figure 3: Active Pre-Draw Betting Street displaying dealt cards on the 3D felt, a 10-Soul LIFO Pot Stack, player betting controls (Check, Bet, Bluff Raise, Pledge Soul), and the Star Platinum Kinetic Vision scan report popup.

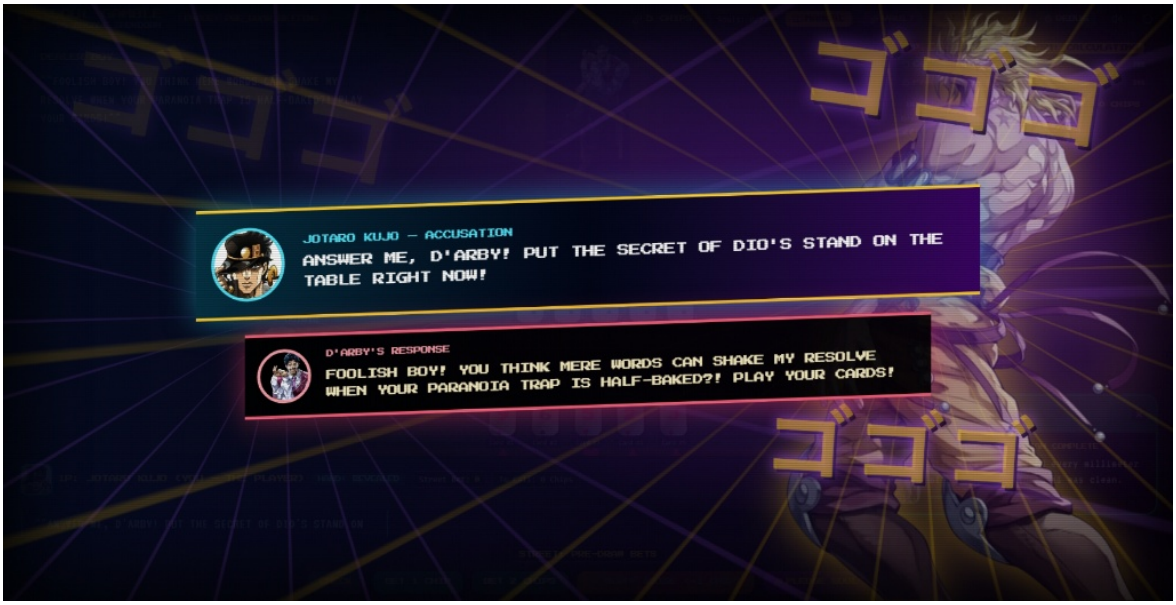


Figure 4: Full-screen dramatic anime confrontation cut-in triggered when demanding the secret of DIO's Stand, where success probability equals D'Arby's current Paranoia percentage.

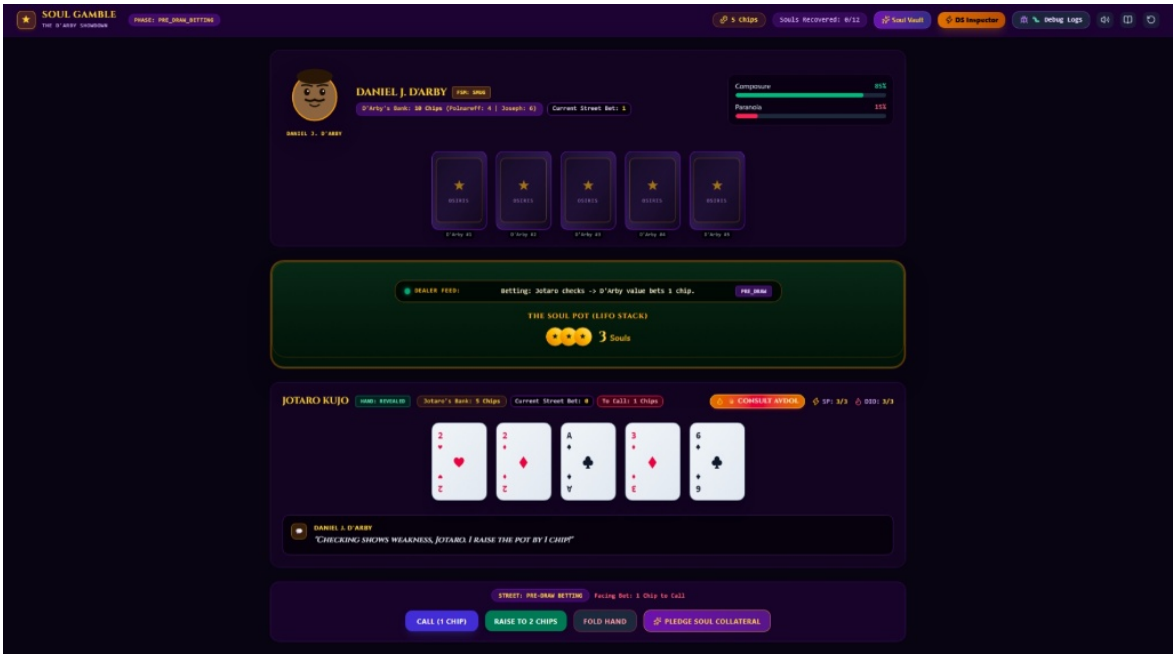


Figure 5: Early prototype view demonstrating 1-to-1 street bet matching, Dealer Feed action logging, and LIFO Soul Pot accumulation during Pre-Draw value betting.